

Yuhui Hong

Department of Genome Sciences
University of Washington
William H. Foegen Hall, 3720 15th Ave NE
Seattle, WA 98195

✉ yuhhong@uw.edu
👤 josiehong.github.io
🔗 github.com/JosieHong
📄 Yuhui Hong
🆔 0000-0002-5647-9714

RESEARCH INTERESTS

My research develops **machine learning that connects molecular structure with the measurements used to characterize it**, addressing open problems from molecule identification to real-world biomolecular discovery. I have developed and validated these methods primarily through mass spectrometry-based molecule identification, with particular attention to reliability under label scarcity and to calibrating prediction confidence. My long-term goal is machine learning that closes the loop between measurement and molecule, extending from metabolite and drug discovery toward broader high-stakes problems in medicine and biology, where predictions must remain grounded in physical and biological constraints to be trusted.

EDUCATION AND APPOINTMENTS

University of Washington	Seattle, WA, US
Postdoctoral Scholar	Aug. 2025 – Present
Advisor: Professor William Stafford Noble	
Area: Computational biology, Deep Learning, Proteomics mass spectrometry	

Indiana University Bloomington	Bloomington, IN, US
Ph.D. in Computer Science	Sep. 2020 – Jul. 2025
Advisor: Prof. Haixu Tang	
Dissertation: “Machine Learning Methods for Small Molecule Analysis in LC-MS/MS”	

Xidian University	Xi'an, Shaanxi, China
B.S. in Computer Science and Technology	Sep. 2015 – Jul. 2019

PUBLICATIONS

Research Highlights. My work includes two papers in *Nature Communications* and an invited review in the *Annual Review of Analytical Chemistry*. I build open-source machine learning systems with real-world adoption, downloaded tens of thousands of times and deployed as public services on Koina and an emerging agentic AI system on GNPS. My models have also been evaluated internally and positively received by biotech, pharmaceutical, and agricultural leaders including AbbVie, Agilent, Amgen, Merck, and Corteva.

PEER-REVIEWED ARTICLES (FIRST AUTHOR)

- 1 **Yuhui Hong**, Sujun Li, Yuzhen Ye, & Haixu Tang (2025). “FIDDLE: a deep learning method for chemical formulas prediction from tandem mass spectra.” *Nature Communications*, 16(1), 11102. [\[Code\]](#) [\[PyPI package\]](#)
Method: A chemically constrained deep-learning model that predicts molecular formulas from tandem mass spectra without a spectral library.

Impact: Released as the open-source msfiddle package (>2,900 downloads); used to discover previously uncharacterized drug metabolites in untargeted metabolomics, and integrated into an agentic AI system for structural elucidation on GNPS.

- 2 **Yuhui Hong**, Yuzhen Ye & Haixu Tang (2025). “Machine Learning in Small-Molecule Mass Spectrometry.” *Annual Review of Analytical Chemistry*, 18.

Impact: Invited review synthesizing the methods, data bottlenecks, and open challenges of machine learning in small-molecule mass spectrometry.

- 3 **Yuhui Hong**, Christopher J Welch, Patrick Piras, & Haixu Tang (2024). “Enhanced Structure-Based Prediction of Chiral Stationary Phases for Chromatographic Enantioseparation from 3D Molecular Conformations.” *Analytical Chemistry*, 96(6), 2351-2359. [\[Code\]](#)

Method: A transfer-learning model that predicts chiral stationary phases for enantiomer separation from 3D molecular conformations.

Impact: Assisted in trial-and-error column selection in chiral method development; developed with industry chromatography experts.

- 4 **Yuhui Hong***, Mahsa Monshizadeh*, & Yuzhen Ye (2024). “Multitask Knowledge-primed Neural Network for Predicting Missing Metadata and Host Phenotype based on Human Microbiome.” *Bioinformatics Advances*, vbae203. [\[Code\]](#)

Method: A multitask, knowledge-primed neural network that jointly predicts host phenotype and imputes missing metadata (e.g., age, sex) from microbiome composition, using auxiliary decoders when metadata is unavailable.

Impact: Applied to microbiome data spanning 25 diseases in mBodyMap; incorporating real or predicted metadata improved both prediction accuracy and generalizability across diseases.

- 5 **Yuhui Hong**, Sujun Li, Christopher J Welch, Shane Tichy, Yuzhen Ye, & Haixu Tang (2023). “3DMolMS: Prediction of Tandem Mass Spectra from Three Dimensional Molecular Conformations.” *Bioinformatics*, btad354. [\[Code\]](#) [\[PyPI package\]](#) [\[Service on Koina\]](#)

Method: A geometric deep-learning model that predicts tandem mass spectra directly from 3D molecular conformations.

Impact: Released as the molnetpack package (>17,000 downloads) and deployed as a public service on Koina, giving the community in silico reference spectra for compounds without experimental library entries.

PEER-REVIEWED ARTICLES (COLLABORATIVE AUTHOR)

- 6 Ludwig Lautenbacher, Kevin L. Yang, Tobias Kockmann, Christian Panse, Wassim Gabriel, Dulguun Bold, Elias Kahl, Matthew Chambers, Brendan X. MacLean, Kai Li, Fengchao Yu, Brian C. Searle, Damien Beau Wilburn, Mohammad Reza Zare Shahneh, **Yuhui Hong**, Haixu Tang, Mingxun Wang, Ralf Gabriels, Robbin Bouwmeester, Robbe Devreese, Jesse Angelis, Eduard Sabidó, Tobias K. Schmidt, Alexey I. Nesvizhskii, & Mathias Wilhelm (2025). “Koina: Democratizing machine learning for proteomics research.” *Nature Communications*, 16(1), 9933. [\[Website\]](#) [\[Code\]](#)

- 7 Yaochen Li, **Yuhui Hong**, Yonghong Song, Chao Zhu, Ying Zhang, & Ruihao Wang (2022). “SiamPolar: Semi-supervised Realtime Video Object Segmentation with Polar Representation.” *Neurocomputing*, 467, 491-503. [\[Code\]](#)

- 8 Yaochen Li, Chao Zhu, Yuehu Liu, **Yuhui Hong**, & Jianji Wang (2021). “Geometric and Semantic Analysis of Road Image Sequences for Traffic Scene Construction.” *Neurocomputing*, 465, 336-349. [\[Code\]](#)

BOOKS & PATENTS

- 1 Qingyang Xiao, Kaiyuan Liu, **Yuhui Hong** & Haixu Tang (2024). “Neural Networks for Chemists.” *American Chemical Society*, DOI:10.1021/acsinfocus.7e8012. [\[Primer\]](#)
- 2 Haixu Tang, **Yuhui Hong**, & Sujun Li (2025). “Method of predicting MS/MS spectra and properties of chemical compounds.” US Patent No. US20250356958A1. [\[Patent\]](#)

ONGOING ARTICLES

- 1 Xianghu Wang, Abubaker Patan, Haoqi Nina Zhao, Vincent Charron-Lamoureux, Daniel Petras, **Yuhui Hong**, Benjamin P. Bowen, Trent Northen, Pieter C. Dorrestein, Mingxun Wang (2026). “Agentic AI for Structural Elucidation and Discovery of Drug Metabolites from Mass Spectrometry Data.” (Under review). [\[Preprint\]](#)
- 2 **Yuhui Hong**, & Haixu Tang (2025). “A Task-Specific Transfer Learning Approach to Enhancing Small Molecule Retention Time Prediction with Limited Data.” (Under review). [\[Preprint\]](#) [\[Code\]](#)
- 3 **Yuhui Hong**^{*}, Mahsa Monshizadeh^{*}, & Yuzhen Ye (2025). “Confounder Free Predictive Models for Microbiome-based Host Phenotype Prediction.” (Under review). [\[Preprint\]](#) [\[Code\]](#)

^{*} Equal contribution as co-first authors.

CONFERENCE PRESENTATIONS

- 1 “De novo sequencing of chimeric spectra using Casanovo.” *74th Conference on Mass Spectrometry and Allied Topics*. May 31 – Jun. 4, 2026. San Diego, CA. (**Oral rate: ~14%**). [\[Slides\]](#)
- 2 “A Task-Specific Transfer Learning Approach to Enhancing Small Molecule Retention Time Prediction with Limited Data.” *73rd Conference on Mass Spectrometry and Allied Topics*. Jun. 1–5, 2025. Baltimore, MD. (**Oral rate: 13.9%**). [\[Slides\]](#)
- 3 “Predicting Compositional Fragments of Compounds from Their Tandem Mass Spectra Using Deep Neural Networks.” *72nd Conference on Mass Spectrometry and Allied Topics*. Jun. 2–6, 2024. Anaheim, CA. (Poster presentation). [\[Poster\]](#)
- 4 “A Machine Learning Model for Chemical Formula Prediction Using Tandem Mass Spectra of Compounds.” *71st Conference on Mass Spectrometry and Allied Topics*. Jun. 4–8, 2023. Houston, TX. (**Oral rate: 10.7%**). [\[Slides\]](#)
- 5 “Prediction of Molecular Tandem Mass Spectra Using 3-Dimensional Conformers.” *70th Conference on Mass Spectrometry and Allied Topics*. Jun. 5–9, 2022. Minneapolis, MN. (Poster presentation). [\[Poster\]](#)

INVITED TALKS

- 1 **Invited talk**. “Beyond database search: machine learning method for small molecule identification.” *College of Science, San Diego State University*. Feb. 11, 2026. San Diego, CA.
- 2 **Invited talk**. “Enhanced Structure-Based Prediction of Chiral Stationary Phases for Chromatographic Enantioseparation from 3D Molecular Conformations.” *Research Group, Amgen*. Oct. 11, 2024. Virtual.

TEACHING EXPERIENCE

Course Designer DSCI-D590, AI on Ramp	Indiana University Bloomington Fall 2025 – Present
Instructor DSCI-D590, Topics in Data Science	Indiana University Bloomington Spring 2025
Instructor INFO-I529, Machine Learning Bioinformatics	Indiana University Bloomington Fall 2024
Assistant Instructor DSCI-D351, Big Data Analytics (Instructor: Prof. Haixu Tang)	Indiana University Bloomington Aug. – Sep. 2024

SCHOLARSHIPS AND AWARDS

- **UW Data Science Fellow at the eScience Institute** 2025
University of Washington
Awarded \$1,000 fellowship for mass spectra data analysis
- **Luddy Outstanding Research Award** 2025
Indiana University Bloomington
Sole recipient university-wide for outstanding research contributions
- **Meritorious Winner of MCM (Mathematical Contest In Modeling)** 2018
Consortium for Mathematics and Its Application (COMAP)
Top 10% internationally (ranked among 8,085 participating teams)

PROFESSIONAL SERVICES

- **Reviewer:** (conferences) KDD 2026, 2027, ACM BCB 2025, 2026, EMBC 2026; (journals) Journal of Chemical Information and Modeling, Journal of Chromatography A, BMC Genomics, BMC Bioinformatics, IEEE Transactions on Computational Biology and Bioinformatics, PeerJ Computer Science, Pharmaceutical Research, Beilstein Journal of Organic Chemistry, Chemical Physics Letters.
- **Sub-reviewer:** (conferences) RECOMB 2022, 2023, 2025, 2026, ISMB/ECCB 2023, 2025, ACM BCB 2024; (journals) Analytical Chemistry, International Journal of Mass Spectrometry.
assisted in reviewing papers under the guidance of Professor William Noble and Professor Haixu Tang

Last updated: July 3, 2026